



IIT JEE | MEDICAL | FOUNDATION

illuminati - 2020

Advanced Physics Assignment-1

Electrostatics, DC Circuits, Capacitors, Magnetism, EMI, AC Circuits

SECTION-1

SINGLE CORRECT ANSWER TYPE

- 1.(A) For the first released charge,

$$\text{Initial potential energy} = q(V_1 + V_2 + \dots + V_{n-1}) = \text{final kinetic energy } k_1$$

For the second released charge,

$$\text{Initial potential energy} = q(V_1 + V_2 + \dots + V_{n-2}) = \text{final kinetic energy } k_2$$

$$k = k_1 - k_2 = q(V_{n-1}) = q \frac{1}{4\pi\epsilon_0} \frac{q}{a} \quad \therefore \quad q = \sqrt{4\pi\epsilon_0 ak}$$

- 2.(A) Let
- I
- be the current.

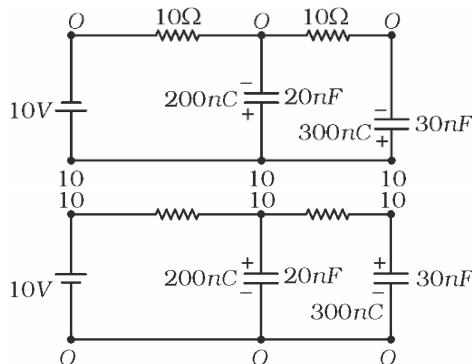
$$\therefore \quad \text{Voltage across resistor} = \sqrt{I}$$

$$\text{Applying KVL : } 1.5 - 0.5I - \sqrt{I} = 0$$

$$\Rightarrow (0.5\sqrt{I} - 0.5)(\sqrt{I} + 3) = 0 \quad \therefore \quad \sqrt{I} = 1 \Rightarrow I = 1A$$

$$\text{Voltage across cell} = 1.5 - 0.5I = 1 \text{ volt}$$

- 3.(A) Initial charges :



Final charges :

$$\begin{aligned} \text{Charge supplied by cell} &= 500nC - (-500nC) \\ &= 1000nC = 1\mu C \end{aligned}$$

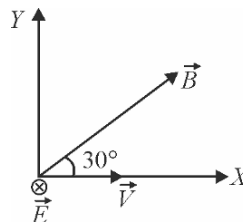
$$\text{As } \Delta U_c = 0 \quad \therefore \quad H_{\text{loss}} = w_{\text{cell}} = (10V)(1\mu C) = 10\mu J$$

- 4.(C)
- $qE = qvB \sin 30^\circ$

$$\Rightarrow v = \frac{5000}{40 \times 10^{-3} \times 0.5} = 2.5 \times 10^5 \text{ m/s}$$

After electric field is removed,

$$\begin{aligned} P &= (v \cos 30^\circ) \left(\frac{2\pi m}{Bq} \right) \\ &= 2.5 \times 10^5 \times \frac{\sqrt{3}}{2} \times \frac{2\pi \times 3 \times 10^{-19}}{40 \times 10^{-3} \times 2 \times 10^{-12}} \approx 5m \end{aligned}$$



5.(B) Induced emf in complete loop of radius $3R$ is

$$\varepsilon_{ind} = -\frac{d\phi}{dt} = -[\pi R^2 2K - 3\pi R^2 4K] = 10\pi R^2 K \quad \therefore \quad \varepsilon_{AB} = 10\pi R^2 K \left(\frac{\theta}{2\pi} \right) = 5R^2 K \theta$$

6.(A) $X_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 200 \times 10^{-6}} = \frac{50}{\pi} \Omega$

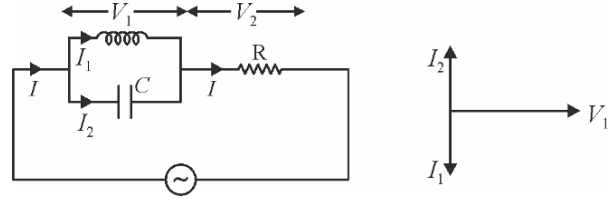
$$X_L = \omega L = 100\pi \times \frac{50}{\pi^2} \times 10^{-2} = \frac{50}{\pi} \Omega$$

$$I_1 = \frac{V_1}{X_C}, \quad I_2 = \frac{V_1}{X_C}$$

As $X_L = X_C \Rightarrow I_1 = I_2$

$$\therefore I = 0, V_2 = 0$$

$$v_1 = v = 5 \sin 100\pi t$$



7.(B) Initial reading of Ammeter $= \frac{\varepsilon}{R_A + R_V}$

Initial reading of voltmeter $= \frac{\varepsilon}{R_A + R_V} R_V$

Final reading of Ammeter $= \eta \left(\frac{\varepsilon}{R_A + R_V} \right)$

Final reading of voltmeter $= \frac{1}{\eta} \left(\frac{\varepsilon R_V}{R_A + R_V} \right)$

$$\therefore \frac{\eta \varepsilon}{R_A + R_V} = \frac{\varepsilon}{R_A + \frac{R_V R}{R_V + R}} \quad \dots (i)$$

$$\text{and} \quad \left(\frac{\eta \varepsilon}{R_A + R_V} \right) \left(\frac{R R_V}{R_V + R} \right) = \frac{1}{\eta} \left(\frac{\varepsilon R_V}{R_A + R_V} \right) \quad \dots (ii)$$

from (ii) : $\frac{R R_V}{R_V + R} = \frac{R_V}{\eta^2}$

putting in (i) :

$$\frac{\eta \varepsilon}{R_A + R_V} = \frac{\varepsilon}{R_A + \frac{R_V}{\eta^2}} = \frac{\eta^2 \varepsilon}{\eta^2 R_A + R_V}$$

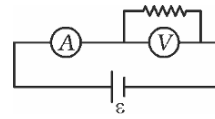
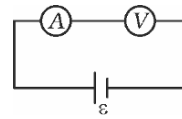
$$\Rightarrow \eta^2 R_A + R_V = \eta R_A + \eta R_V \Rightarrow \eta R_A (\eta - 1) = R_V (\eta - 1) \Rightarrow R_V = \eta R_A$$

Initial reading of voltmeter $= \frac{\varepsilon R_V}{R_A + R_V} = \frac{\eta \varepsilon}{\eta + 1}$

8.(D) $V = x^2 - y^2$

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} = -2x \hat{i} + 2y \hat{j}$$

Slope of tangent to field lines $= \frac{E_y}{E_x} = -\frac{y}{x}$

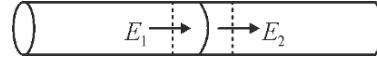


$$9.(C) \quad J_1 = \frac{I}{A} = \frac{I}{\rho_1} E_1 \Rightarrow E_1 = \rho_1 \frac{I}{A}$$

$$J_2 = \frac{I}{A} = \frac{1}{\rho_2} E_2 \Rightarrow E_2 = \rho_2 \frac{I}{A}$$

For a volume element enclosing the junction,

$$\phi = E_2 A - E_1 A = \frac{Q_{enc}}{\epsilon_0}$$



$$\Rightarrow Q_{enc} = \epsilon_0 A(E_2 - E_1) = \epsilon_0 I(\rho_2 - \rho_1)$$

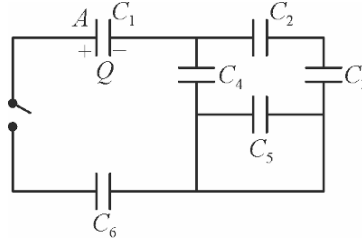
10.(B) C_5 is short-circuited

Equivalent capacitance of

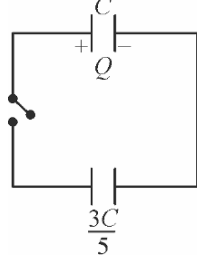
$$C_1, C_3, C_4 \text{ is } \frac{C}{2} + C = \frac{3C}{2}$$

This is in series with C_6

$$\therefore C_{eq} = \frac{\frac{3C}{2} C}{\frac{3C}{2} + C} = \frac{3C}{5}$$



$\therefore$ The effective circuit is



$$\therefore CV + \frac{3C}{5}V = Q$$

$$\text{Also } CV = 5\mu C$$

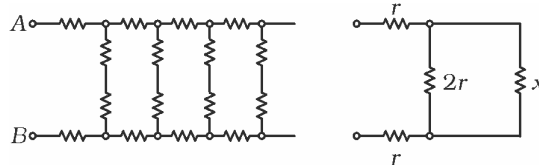
$$\therefore 5 + \frac{3}{5}(5) = Q$$

$$\Rightarrow Q = 8\mu C$$

$$11.(B) \quad \phi = -B \frac{\pi a^2}{2} + B \frac{\pi a^2}{2} \sin 30^\circ = -\frac{B\pi a^2}{4}$$

$$\therefore \theta = \frac{|\Delta\phi|}{R} = (B_0 t - 0) \frac{\pi a^2}{4} \frac{1}{2\pi a r_0} = \frac{B_0 a t}{8r_0}$$

12.(C) Due to symmetry, there will be no current in the resistances lying on the line of symmetry. So, circuit is equivalent to



$$\frac{2rx}{2r+x} + 2r = x \Rightarrow 2rx + 4r^2 + 2rx = 2rx + x^2$$

$$\Rightarrow x^2 - 2rx - 4r^2 = 0 \quad \therefore x = (\sqrt{5} + 1)r$$

$$13.(C) \quad V_1 = V_0 \left(1 - e^{\frac{-t}{RC}} \right)$$

$$\Rightarrow V_0 - V_1 = V_0 e^{\frac{-t}{RC}} \quad \therefore t = RC \ln \frac{V_0}{V_0 - V_1}$$

- 14.(A) Capacitance of the system of two spheres in vacuum can be calculated as :

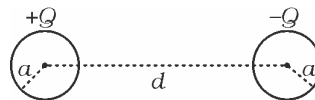
$$V = V_+ - V_- = \frac{kQ}{a} - \frac{k(-Q)}{a} = \frac{Q}{2\pi\epsilon_0 a}$$

$$\Rightarrow Q = 2\pi\epsilon_0 aV \quad \therefore C = 2\pi\epsilon_0 a$$

The given case is the case of a leaky capacitor

$$C = k(2\pi\epsilon_0 a), \text{ where } \epsilon = k\epsilon_0$$

$$\therefore RC = \rho k\epsilon_0 \Rightarrow R = \frac{\rho k\epsilon_0}{k2\pi\epsilon_0 a} = \frac{\rho}{2\pi a}$$



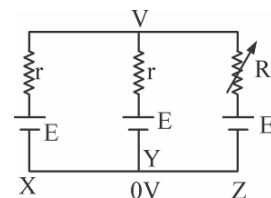
- 15.(C) Redrawing the given circuit diagram as shown below :

Using point potential theory

$$\frac{V-E}{r} + \frac{V-E}{r} + \frac{V-E}{R} = 0 \Rightarrow (V-E) \left(\frac{2}{r} + \frac{1}{R} \right) = 0$$

$$\text{As } \frac{2}{r} + \frac{1}{R} \neq 0 \quad \text{so } V-E=0$$

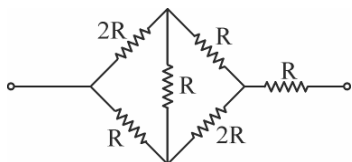
$$\text{So, current through } R, i = \frac{V-E}{R} = 0 \text{ whatever be the value of } R.$$



- 16.(C) Let σ be the charge density of conducting plate and V be the volume of either dielectric

$$\therefore \frac{U_1}{U_2} = \frac{\left(\frac{1}{2} K_1 \epsilon_0 E_1^2 \right) V}{\left(\frac{1}{2} K_2 \epsilon_0 E_2^2 \right) V} = \frac{K_1 \left(\frac{\sigma}{K_1 \epsilon_0} \right)^2}{K_2 \left(\frac{\sigma}{K_2 \epsilon_0} \right)^2} = \frac{K_2}{K_1}$$

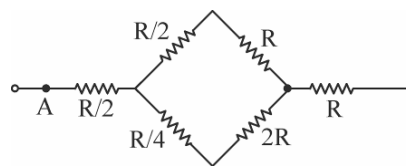
- 17.(D)



From delta – star transformation, the circuit can be redrawn as

$$R_{eq} = \frac{\frac{3R}{2} \times \frac{9R}{4}}{\frac{3R}{2} + \frac{9R}{4}} + R + \frac{R}{2} = \frac{12}{5} R$$

$$I = \frac{V}{2R}; \quad I' = \frac{5V}{12R} \Rightarrow \frac{I'}{I} = \frac{5}{6}$$



- 18.(B) Just before S_1 is closed the potential difference across capacitor 2 is $2E$.

Just after S_1 is closed the potential difference across capacitors 1 and 2 are 0 and $2E$ respectively.

Applying KVL to loop ABCD immediately after S_1 is closed.

$$E = -iR_1 + 0 + 2E \quad \text{or} \quad i = \frac{E}{R_1} \text{ towards left}$$

- 19.(D) When DC is applied across a solenoid, then Resistance, i.e. $R = \frac{V}{I}$

$$R = \frac{100}{1} = 100\Omega$$

$$\text{When AC is applied, then } z = \frac{V}{I}$$

$$Z = \frac{100}{0.5} = 200\Omega$$

Net impedance,

$$Z = \sqrt{R^2 + X_L^2}$$

$$200 = \sqrt{(100)^2 + X_L^2}$$

$$\Rightarrow (200)^2 = (100)^2 + X_L^2 \Rightarrow 30000 = X_L^2 \Rightarrow X_L = 173.205\Omega$$

Now, $X_L = \omega L$

$$\Rightarrow 173.205 = 2\pi fL \Rightarrow \frac{173.205}{2\pi \times f} = L \Rightarrow \frac{173.205}{2 \times 3.14 \times 50} L$$

$$\Rightarrow 0.55 \Omega = L ; L = 0.55\Omega$$

$$20.(C) I = \frac{V}{r_1 + r_2} = k_1\theta_1 = k_2\theta_2 \Rightarrow \frac{k_1}{k_2} = \frac{\theta_2}{\theta_1}$$

$$I'_1 = \frac{V}{r_1} = k_1\theta'_1, I'_2 = \frac{V}{r_2} = k_2\theta'_2 \Rightarrow \frac{r_2}{r_1} = \frac{k_1\theta'_1}{k_2\theta'_2} = \frac{\theta_2}{\theta_1} \frac{\theta'_1}{\theta'_2}$$

SECTION-2

LINK-COMPREHENSION TYPE

$$21.(B) V_0 = \frac{kq}{r} \Rightarrow r = \frac{kq}{V_0}$$

$$\frac{dr}{dt} = \frac{k}{V_0} \frac{dq}{dt} = \frac{k}{V_0} \left(-\frac{V_0}{R} \right) = -\frac{1}{4\pi\epsilon_0 R}$$

$$22.(C) \frac{a}{2} = a - \frac{1}{4\pi\epsilon_0 R} t \Rightarrow t = 2\pi\epsilon_0 Ra \quad \therefore H = I^2 Rt = \frac{V_0^2}{R^2} R 2\pi\epsilon_0 a = 2\pi\epsilon_0 a V_0^2$$

$$23.(A) V_{OR} = 4, V_{OL} = 3$$

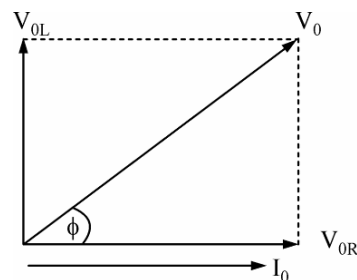
$$\therefore V_0 = \sqrt{3^2 + 4^2} = 5$$

$$\therefore I = I_0 \sin(\omega t - \phi)$$

$$V_R = 4 \sin(\omega t - \phi); V_L = 3 \sin(\omega t - \phi + 90^\circ)$$

$$\text{When } V_R = 2, (\omega t - \phi) = 30^\circ$$

$$\therefore V_L = 3 \sin 120^\circ = 3 \cos 30^\circ$$



$$24.(B) V = V_R + V_L = 2 + \frac{3\sqrt{3}}{2} = \frac{4 + 3\sqrt{3}}{2}$$

$$25.(D) V = V_0 \sin \omega t$$

$$I = I_0 \sin(\omega t - \phi)$$

Depends on time

$$26.(C) f = \frac{1}{P} = \frac{1}{10} = 0.1m = 10cm$$

$$v = 15cm$$

$$\therefore \frac{1}{15} - \frac{1}{u} = \frac{1}{10} \Rightarrow u = -30cm$$

$$\text{Vel. Of image} = \frac{v}{u} (\text{Vel of object}) = \frac{15}{-30}(100) = -50\text{m/s}$$

$$\text{Distance moved in } 2\text{ms} = 50(2 \times 10^{-3}) = 0.1\text{m} = 10\text{cm}$$

27.(D) After 1 time constant, $Q = 0.37Q_0$

$$\begin{aligned} \therefore \text{Energy delivered} &= \frac{Q_0^2}{2C} - \frac{(0.37Q_0)^2}{2C} \\ &= \frac{Q_0^2}{2C}(1 - 0.137) = 4(0.863) \\ &= 3.45J \end{aligned}$$

28.(A) $\tau = RC \Rightarrow R = \frac{\tau}{C} = \frac{0.2 \times 10^{-3}}{200 \times 10^{-6}} = 1\Omega$

$$U_0 = \frac{1}{2}CV_0^2 \Rightarrow V_0^2 = \frac{2(4)}{200 \times 10^{-6}} = 4 \times 10^4$$

$$\therefore V_0 = 200 \text{ volts} \quad \therefore I_0 = \frac{V_0}{R} = 200A$$

29.(B) $i_2 = i_3 = I_b$

$$V_1 = (V_2 + V_3) = V \text{ and } P_2 = P_3$$

$$\Rightarrow R_2 = R_3 \text{ and } V_2 = V_3 = V/2$$

$$P_1 = P_2 = P_3 = P_4 \text{ (given)}$$

$$P_1 = \frac{V^2}{36}; P_3 = \frac{(V/2)^2}{R_3}$$

$$\text{As } P_1 = P_3 \Rightarrow R_3 = 9\Omega$$

30.(A) $I_a = \frac{I}{3}$ and $I_4 = I$

$$P_1 = P_4 = 4W$$

$$\left(\frac{I}{3}\right)^2 R_1 = (I)^2 R_4 = 4W$$

$$\Rightarrow \frac{R_1}{9} = R_4 \Rightarrow 4\Omega = R_4$$

31.(B) Also $I = 1A$

$$R_{eq} = 16\Omega \text{ and } I = 1A$$

$$\varepsilon = 16V$$

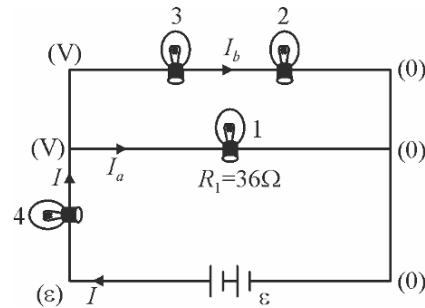
32.(C) 33.(D) 34.(C)

For $t = 0$ to $t_0 = RC$ seconds, the circuit is of charging type. The charging equation for this time is

$$q = CE \left(1 - e^{-\frac{t}{RC}} \right)$$

$$\text{Therefore the charge on capacitor at time } t_0 = RC \text{ is } q_0 = CE \left(1 - \frac{1}{e} \right)$$

For $t = RC$ to $t = 2RC$ seconds, the circuit is of discharging type. The charge and current equation for this time are



$$q = q_0 e^{-\frac{t-t_0}{RC}} \text{ and } i = \frac{q_0}{RC} e^{-\frac{t-t_0}{RC}}$$

Hence charge at $t = 2RC$ and current at $t = 1.5RC$ are

$$q = q_0 e^{-\frac{2RC-RC}{RC}} = \frac{q_0}{e} = \frac{1}{e} CE \left(1 - \frac{1}{e}\right)$$

$$\text{and } i = \frac{q_0}{RC} e^{-\frac{1.5RC-RC}{RC}} = \frac{q_0}{\sqrt{e}RC} = \frac{E}{\sqrt{e}R} \left(1 - \frac{1}{e}\right) \text{ respectively}$$

Since the capacitor gets more charged up from $t = 2RC$ to $t = 3RC$ than in the interval $t = 0$ to $t = RC$, the graph representing the charge variation is as shown in figure.



35.(B) The force per unit length f between two indefinite parallel straight wires separated by a distance h is.

$$f = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{h}$$

For $I_1 = I_2 = I$ and length $2\pi a$, the force F induced on C_2 by the neighbor coils C_1 is

$$F = \frac{\mu_0 a}{h} I^2$$

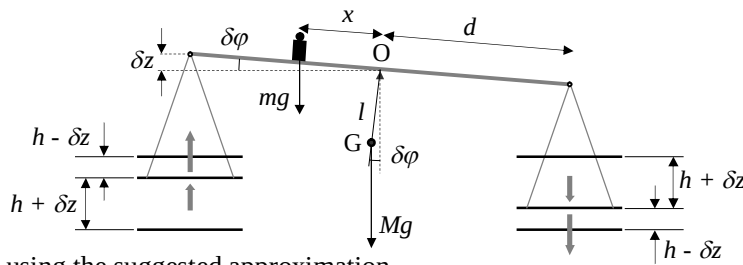
36.(A) In equilibrium $m g x = 4 F d$

$$\text{Then } m g x = \frac{4\mu_0 a d}{h} I^2 \quad (1)$$

$$\text{So that } I = \left(\frac{m g h x}{4\mu_0 a d} \right)^{1/2}$$

37.(B) The balance comes back towards the equilibrium position for a little angular deviation $\delta\phi$ if the gravity torques with respect to the fulcrum O are greater than the magnetic torques.

$$Mg l \sin\delta\phi + m g x \cos\delta\phi > 2\mu_0 a I^2 \left(\frac{1}{h-\delta z} + \frac{1}{h+\delta z} \right) d \cos\delta\phi$$



Therefore, using the suggested approximation

$$Mg l \sin\delta\phi + m g x \cos\delta\phi > \frac{4\mu_0 a d I^2}{h} \left(1 + \frac{\delta z^2}{h^2} \right) \cos\delta\phi$$

Taking into account the equilibrium condition (1), one obtains

$$Mg l \sin\delta\phi > m g x \frac{\delta z^2}{h^2} \cos\delta\phi$$

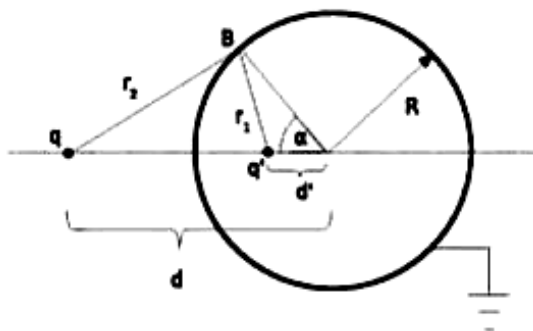
$$\text{Finally, for } \tan \delta\phi \approx \sin \delta\phi = \frac{\delta z}{d}$$

$$\delta z < \frac{M l h^2}{m x d} \Rightarrow \delta z_{\max} = \frac{M l h^2}{m x d}$$

38-39. 38.(C) 39.(B) 40.(A)

Sphere is grounded, its potential vanishes, $V = 0$.

Let us consider an arbitrary point B on the surface of the sphere as depicted in figure.



The distance of point B from the charge q' is $r_1 = \sqrt{R^2 + d'^2 - 2Rd' \cos \alpha}$ (1)

Whereas the distance of the point B from the charge q is given with the expression

$$r_2 = \sqrt{R^2 + d^2 - 2Rd \cos \alpha} \quad (2)$$

The electric potential at the point B is $V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_2} + \frac{q'}{r_1} \right)$

This potential must vanish, i.e. its numerical value is 0 V.

$$\frac{q}{r_2} + \frac{q'}{r_1} = 0 \quad (3)$$

Combining (1), (2) and (3) we obtain $R^2 + d^2 - 2Rd \cos \alpha = \left(\frac{q}{q'} \right)^2 (R^2 + d'^2 - 2Rd' \cos \alpha)$

As the surface of the sphere must be equipotential, this condition must be satisfied for every angle α what leads to the following results

$$d^2 + R^2 = \left(\frac{q}{q'} \right)^2 (R^2 + d'^2) \quad \text{and} \quad dR = \left(\frac{q}{q'} \right)^2 (d'R)$$

By solving, we obtain the expression for the distance d' of the charge q' from the center of the sphere $d' = \frac{R^2}{d}$ and the size of the charge q' ; $q' = -q \frac{R}{d}$

Finally, the magnitude of force acting on the charge q is $F = \frac{1}{4\pi\epsilon_0} \frac{q^2 R d}{(d^2 - R^2)^2}$

The force is apparently attractive.

SECTION-3

MULTIPLE CORRECT ANSWER TYPE

$$41.(BD) \quad X_C = \frac{1}{100\pi \times \frac{20}{\pi} \times 10^{-6}} = 500 \Omega$$

$$X_L = 100\pi \times \frac{5}{\pi} = 500 \Omega$$

As $X_C = X_L \Rightarrow$ Voltage across the combination is zero.

So, the resistor is short circuited and as $X_L(30 \Omega) > X_C(20 \Omega)$

$\therefore$ Circuit is purely inductive.

- 42.(AD) While revolving in conical pendulum force equations on the charge can be written as

$$T \sin \theta + F = m\omega^2 (L \sin \theta)$$

$$T \sin \theta = mg \quad \Rightarrow \quad \frac{mg}{\cos \theta} \sin \theta + \theta(\omega L \sin \theta)B = m\omega^2 L \sin \theta$$

$$\Rightarrow B = \frac{m\omega^2 L \sin \theta}{(\omega L \sin \theta)q} - \frac{mg \sin \theta}{\cos \theta} \Rightarrow B = \frac{m\omega}{q} - \frac{mg}{q\omega L \cos \theta} = \frac{1}{\beta} \left[\omega - \frac{g}{\omega L \cos \theta} \right]$$

- 43.(ABD) Total charge = $\int I dt$ = Area under the curve = 10 C hence Change in magnetic flux
= $R \times Q = 100 \text{ Wb}$.

As current is decreasing, EMF induced is decreasing. Average current = $\frac{\int I dt}{\int dt} = 5 \text{ A}$

$$\text{Total heat produced} = \int I^2 R dt = \int_0^2 (-5t + 10)^2 \cdot 10 \cdot dt = \frac{200}{3} \text{ J}$$

$$\text{Maximum Power} = I^2 R \text{ when } I \text{ is maximum current.} = 100 \times 1 = 100 \text{ W}$$

- 44.(ABC) When S_1 is closed, only C is charged.

$$\text{Charge through the battery } Q = CV = 120 \mu\text{C}$$

As S_1 is opened and S_2 is closed, charge is redistributed between C_1 and C_2 . Let these be Q_1 and Q_2 .

$$Q_1 + Q_2 = 120 \mu\text{C} \quad \dots\dots\dots (1)$$

Using KVL in closed loop

$$\frac{Q_1}{C_1} - \frac{Q_2}{C_2} = 0 \quad \Rightarrow \quad \frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$\text{Solving } Q_1 = 80 \mu\text{C} \text{ and } Q_2 = 40 \mu\text{C}$$

$$\Delta H = \left[\frac{Q^2}{2C_1} \right] - \left[\frac{Q_1^2}{2C_1} + \frac{Q_2^2}{2C_2} \right] = 400 \times 10^{-6} \text{ J}$$

- 45.(AD) At R, $E_{net} = E + \frac{Q}{2A\epsilon_0} - \frac{Q}{2A\epsilon_0} = E$

$$\text{At S, } E_{net} = E + \frac{Q}{2A\epsilon_0} + \frac{Q}{2A\epsilon_0} = E + \frac{Q}{A\epsilon_0}$$

$$\text{At T, } E_{net} = E + \frac{Q}{2A\epsilon_0} - \frac{Q}{2A\epsilon_0} = E$$

- 46.(ACD) Only B is wrong as it is material property. As I is $I \propto j \propto v_d$ hence A, C & D correct

- 47.(ABC) When switch is closed capacitor behaves as conductor hence A is correct.

At steady state and finally behaves like a very large resistance hence current in Ammeter will be $V_0 / R_1 + R_2$ hence B is correct. Just after opening switch capacitor starts discharging through R_2 hence initial current will be $V_0 / 3R_2$ hence C is correct.

- 48.(ACD) $OA = OC = L$

$$OP = \frac{\sqrt{3}L}{2}$$

$$V_A - V_0 = \frac{\omega BL^2}{2}$$

$$V_A - V_C = 0$$

$$V_C - V_0 = \frac{\omega BL^2}{2}$$

$$V_P - V_0 = \frac{\omega B}{2} \left(\frac{3L^2}{4} \right)$$

$$= \frac{3\omega BL^2}{8}$$

$$V_C - V_P = (V_C - V_0) - (V_P - V_0) = \frac{\omega BL^2}{2} - \frac{3\omega BL^2}{8} = \frac{\omega BL^2}{8}$$

49.(AC) $\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$

$$V = xy^2 - 4x^3$$

$$E_x = -\frac{\partial V}{\partial x} = -[y^2 - 12x^2]$$

$$E_x \text{ at } (1,1) = 11 \text{ V/m}$$

$$E_y = -\frac{\partial V}{\partial y} = -[2xy]$$

$$E_y \text{ at } (1,1) = -2 \text{ volt/m}$$

50.(AD)

Force experience by rod is

$$\Rightarrow F = (\lambda \ell) E \quad \dots (1)$$

$$F = ma \quad \dots (2)$$

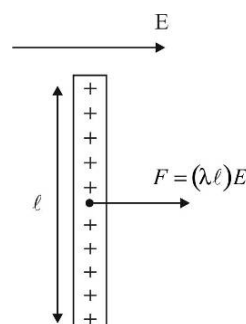
Form equation (1) and (2)

$$(\lambda \ell) E = ma$$

$$\Rightarrow a = \frac{\lambda E}{m / \ell} \quad \Rightarrow a = \frac{\lambda E}{\mu} \quad \dots (3)$$

Now $v^2 = u^2 + 2aS$

$$v^2 = (0)^2 + 2 \times \frac{\lambda E}{\mu} D \quad \Rightarrow v = \sqrt{\frac{2\lambda ED}{\mu}}$$



51.(ABC) Due to electrostatic shielding potential at any outside point due to inner charge will be zero.

52.(ABC) From given conditions,

$$\Rightarrow V_B = \frac{k(Q - q_1)}{3a} + \frac{kq_2}{2a} + \frac{kq_1}{2a} = 0$$

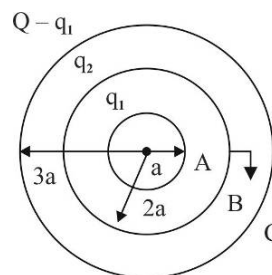
$$\Rightarrow 2Q + q_1 + 3q_2 = 0 \quad \dots (1)$$

Using $V_A = V_C$

$$\frac{k(Q - q_1)}{3a} + \frac{kq_2}{3a} + \frac{kq_1}{3a} = \frac{kq_1}{a} + \frac{k(Q - q_1)}{3a} + \frac{kq_2}{2a}$$

$$\Rightarrow q_1 = -\frac{q_2}{4} \quad \dots (2)$$

Using it in (1), $q_2 = -\frac{8}{11}Q$.



53.(ABD) As cylinders are equipotential hence B is true. By Gauss thm A is true and C is False. D is true as inner will have zero potential.

54.(AC)

55.(BCD) At a particular instant, by Kirchhoff's law

$$\begin{aligned} V_{\text{source}} &= |V_{\text{inductor}} + V_{\text{resistor}}| \\ &= |\pm 3V + 4V| = 7V \text{ or } 1V \\ i &= \frac{\text{Voltage across resistor}}{\text{Resistance}} = 0.8 \text{ amp} \end{aligned}$$

56.(ACD)

$$P_1 = P_2; Z_1 = Z_2$$

$$\sqrt{(X_L - X_C)^2 + R^2} = \sqrt{X_L^2 + R^2}$$

$$X_C = 2X_L$$

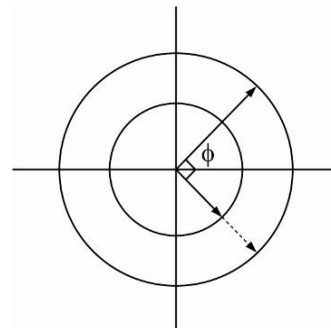
$$\text{and in LCR circuit } \tan \phi = \frac{X_C - X_L}{R}$$

$$\text{In LR circuit } \tan(90^\circ - \phi) = \frac{X_L}{R}$$

$$\therefore X_L = R$$

$$P = \frac{\varepsilon^2}{Z^2} R$$

$$\therefore R = \frac{230 \times 230}{8 \times 2} = 3.3 k\Omega$$



57.(AD)

Potential difference across R_1 increases

$$\therefore i_{R_1} \uparrow$$

Potential difference across R_2 is constant

$$\therefore i_{R_2} \text{ remains unchanged}$$

$$\text{Potential difference across } L = 0 \Rightarrow \frac{L di_L}{dt} = 0$$

$$\Rightarrow i_L = \text{constant}$$

$$i_{\text{battery}} = i_{R_1} + i_{R_2} \Rightarrow \text{increases}$$

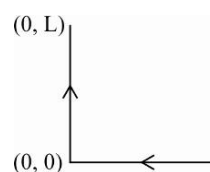
58.(ABD)

$$d\vec{F} = (d\vec{l} \times \vec{B})$$

$$F = \int_0^L I \times dy \times \frac{B_0 y}{L} = \frac{IB_0}{L} \frac{L^2}{2} = \frac{B_0 IL}{2} (\hat{i})$$

$$\int_0^L I \times dy \times \frac{B_0 y}{L} = \frac{IB_0}{L} \frac{L^2}{2} = \frac{B_0 IL}{2} (\hat{i})$$

$B_0 IL$ in (-ve) y-direction.



59.(AC)

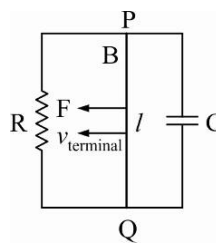
$$\frac{\mu_0 i}{2\pi a} = \frac{\mu_0 J_0}{6} (3a - 2a)$$

$$i = \frac{J_0 \pi a^2}{3}$$

60.(BC)

$$\text{In steady state } Blv_{\text{terminal}} = \frac{q_{\text{max}}}{C}$$

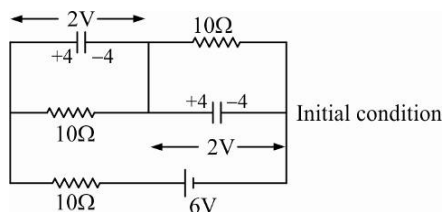
$$\text{Also } iBl = F \text{ and } i = \frac{Blv_{\text{terminal}}}{Bl}$$



SECTION-4

COLUMN MATCHING TYPE

61.(A)



First Step

Dielectric is filled in capacitor C_1

$$Q_2 - Q_1 = 20 - 4 = 16$$

Work done by battery

$$w_{DB} = Q\Delta V$$

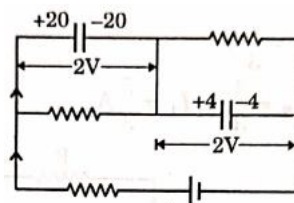
$$w_{AB} = 16 \times 6 = 96$$

Heat generated

$$u_i + w_{DB} = u_f + \text{Heat}$$

$$-16 + 96 = \text{Heat}$$

Heat generated = 80

62.(B) $Q_2 - Q_1 = 20$

Work done by battery

$$w_{DB} = Q\Delta V$$

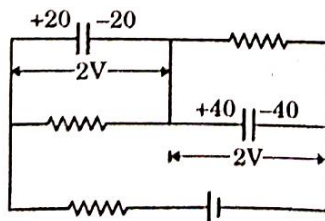
$$w_{AB} = 36 \times 6 = 216$$

Heat generated

$$u_i + w_{DB} = u_f + \text{Heat}$$

$$-36 + 216 = \text{Heat}$$

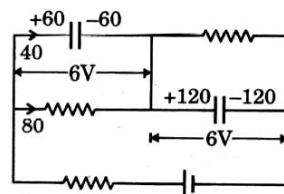
Heat generated = 180

63.(C) $Q_2 - Q_1 = 60$

Work done by battery

$$w_{DB} = Q\Delta V$$

$$w_{AB} = 120 \times 6 = 720$$

Heat generated $u_i + w_{DB} = u_f + \text{Heat}$ 

64. A-S; B-S; C-Q; D-P

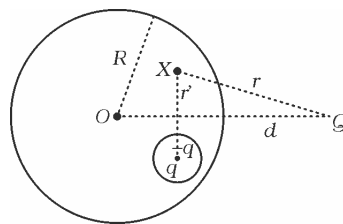
Field and potential due to (q and $-q$) is zero outside cavity surface

$$\therefore V_0 = V_Q + V_{+q} = \frac{kQ}{d} + \frac{kq}{R}$$

Also $V_0 = V_x$ (Conductor is equipotential)As $\vec{E}_{net} = 0$ at X

$$\therefore \vec{E}_{+q} = -\vec{E}_Q \Rightarrow E_{+q} = \frac{kQ}{r^2}$$

$(V_Q + V_{+q})$ is same at all points inside the conductor and equal to that at centre : $\frac{kQ}{d} + \frac{kq}{R}$



$$\therefore V_{+q} \text{ at } X = (V_Q + V_{+q}) - V_Q = \frac{kQ}{d} + \frac{kq}{R} - \frac{kQ}{r}$$

$$V_{-q} \text{ at } x = -\frac{kq}{r'}$$

$$\therefore (V_{+q} + V_{-q}) \text{ at } X = \frac{kQ}{d} + \frac{kq}{R} - \frac{kQ}{r} - \frac{kq}{r'}$$

65. **A-R; B-S; C-P, D-Q** $S = \frac{Q}{P} R$

66. **A-P, Q, R; B-Q, S, T; C-S, T; D-P, R**

$$F = \frac{Q^2}{2A\epsilon_0}, U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}, E = \frac{Q}{A\epsilon_0} = \frac{V}{d}$$

- (A) Q remains same and C decreases (B) Q remains same and C increases
(C) V remains same and C decreases (D) V remains same and C increases

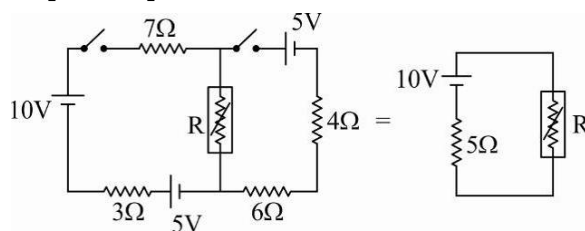
67. **A-Q, S; B-P, R; C-P, R**

From lenz's law, find the direction of induced current

For clockwise current, wire will attract the loop and for anticlockwise current, it will repel the loop.

As the force on all elements of loop is in x-y plane, there is no torque about y-axis.

68. **[A-r] [B-p] [C-q] [D-s]**



So power is maximum when R is 5Ω

For (q) : R is 5Ω when drop in temperature is 20°C so temperature is 30°C .

For (r) : $-\frac{d\theta}{dt} = \frac{\ln 3}{100}(\theta - 20^\circ)$

For (s) : $\int_{50^\circ}^{30^\circ} -\frac{d\theta}{(\theta - 20^\circ)} = \frac{\ln 3}{100} \int_0^t dt$
 $t = 100 \text{ sec}$

By applying loop law we can calculate current when R is 5Ω .

69. **[A-p, q] [B-r, s] [C-r, s] [D-p, q]**

(A) Phase difference $\phi = \tan^{-1}\left(\frac{X}{R}\right) = \frac{\pi}{4}$ for 1 and 2

(B) Phase difference $\phi = \tan^{-1}\left(\frac{X}{R}\right) = \frac{\pi}{4}$ for $1 = -\frac{\pi}{4}$ for 2

(C) Phase difference $\phi = \tan^{-1}\left(\frac{X}{R}\right) = \frac{\pi}{4}$ for $1 = -\frac{\pi}{4}$ for 2

(D) Phase difference $\phi = \tan^{-1}\left(\frac{X}{R}\right) = \frac{\pi}{4}$ for both

70. [A-q, r] [B-p, s] [C-p, r] [D-q, s]

(A) Area $OPMQO = \frac{1}{2} r^2 \theta$

Flux in this area, $\phi_1 = \frac{1}{2} r^2 \theta B$

Induced emf in this area,

$$e_1 = \frac{d\phi_1}{dt} = \frac{1}{2} r^2 \theta \frac{dB}{dt}$$

Area $OPQ = r \sin\left(\frac{\theta}{2}\right) r \cos\left(\frac{\theta}{2}\right) = \frac{1}{2} r^2 \sin \theta$

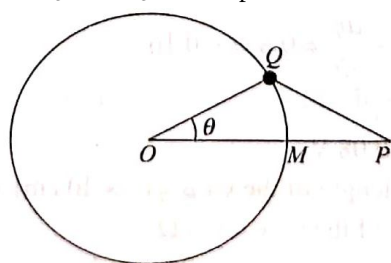
Induced emf in OPQ,

$$e_2 = \frac{d\phi_2}{dt} = \frac{1}{2} r^2 \sin \theta \frac{dB}{dt}$$

e_2 will be only in part PQ, because in OQ and OP induced emf will be zero. Clearly, $e_2 < e_1$, because area $OPQ < OPMQO$.

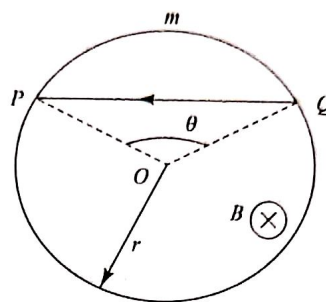
Since B is increasing, so e.m.f. will be in anti-clockwise direction. Hence end P will be positive w.r.t. Q.

- (B) Here emf in OPQ will be due to flux changing in area OMQ . This area is $\frac{1}{2} r^2 \theta$. The entire emf will be in part PQ. End Q will be positive.



- (C) Induced emf $= \frac{1}{2} r^2 \theta \frac{dB}{dt}$. End P will be positive.

- (D) Area in which flux is changing is less than $\frac{1}{2} r^2 \theta$. End Q will be positive.



SECTION-5

NUMERICAL TYPE

71.(0.75) $\phi = \frac{q_{in}}{\epsilon_0} = \frac{\int dq}{\epsilon_0} = \frac{\int A dx \rho}{\epsilon_0} = \frac{A}{\epsilon_0} \int \rho dx = \frac{1}{\epsilon_0}$ [area under the curve] $= \frac{3}{4}$

72.(54) $\phi = \int \vec{E} \cdot d\vec{s}$

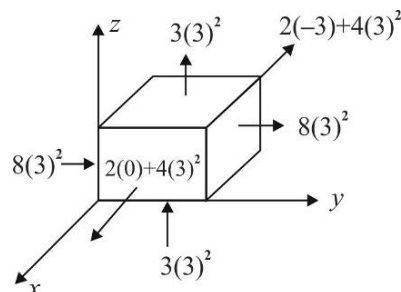
Direction of field at $x = -3$ m is along negative x axis.

Area vector is also along same direction

$$\phi = 6 \times 9 = \frac{Q}{\epsilon_0} \Rightarrow Q = 54 \epsilon_0$$

Components of electric field which are constant, do not contribute in net flux in or out.

$$\frac{q_{in}}{\epsilon_0} = 54 \Rightarrow q_{in} = 54 \epsilon_0$$

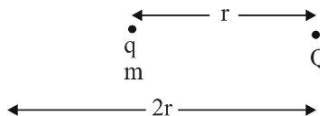


73.(5) Applying conservation of energy, we get

$$\frac{KQq}{r} = \frac{KQq}{2r} + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = \frac{KQq}{2r}$$

$$\therefore v = \sqrt{\frac{KQq}{mr}}$$



Force required to keep Q fixed = Coulombic repulsion between the charges = force on charge q

$$\therefore \frac{1}{2} \text{ Impulse} = \text{change in momentum of } q = mv = \sqrt{\frac{mkQq}{r}} = \sqrt{\frac{Qqm}{4\pi\epsilon_0 r}} \text{ substituting values Ans is 5}$$

74.(4) $\pi R^2 \times \frac{\sigma}{2\epsilon_0} = 8 \times 10^8 \times 2\pi RT$

$$\frac{R}{4\epsilon_0 t} \times \left(\frac{Q}{4\pi R^2} \right)^2 = 8 \times 10^8$$

$$Q = \sqrt{8 \times 10^8 \times 64\pi^2 R^3 \omega t} = \sqrt{\pi} \times \sqrt{\frac{8 \times 10^3 \times 16 \times \frac{1}{8} \times 0.09 \times 10^{-3}}{9 \times 10^9}} = 4\sqrt{\pi} \times 10^{-3}$$

75.(3) $q_1 + q_2 = 0$

$$V_A = \frac{kq_1}{R} + \frac{kQ}{2R} + \frac{kq_2}{4R}$$

$$V_C = \frac{kq_1}{4R} + \frac{kQ}{4R} + \frac{kq_2}{4R}$$

$$\Rightarrow V_A = V_C \Rightarrow q_1 = -Q/3 \text{ and } q_2 = -Q/3$$

76.(1) Minor arc subtends 120° at C_1 & C_2 respectively.

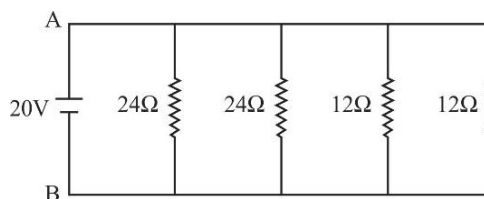
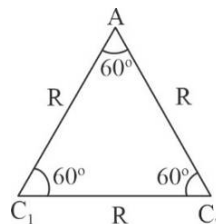
$$R_{\text{minor}} = \frac{1}{3}(36) = 12\Omega$$

$$R_{\text{major}} = \frac{2}{3}(36) = 24\Omega$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{24} + \frac{1}{24} + \frac{1}{12} + \frac{1}{12}$$

$$R_{\text{eq}} = 4\Omega$$

$$P_{\text{cell}} = \frac{\epsilon^2}{R_{\text{eq}}} = \frac{20^2}{4} = 100 = 1 \times 10^2 \text{ watt}$$



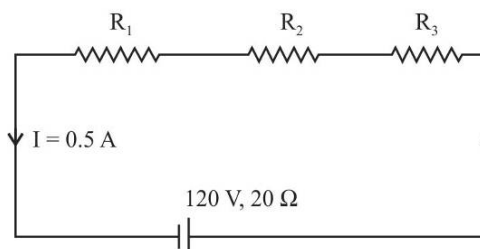
77.(8) $V_1 + V_2 + V_3 = 120 - 20(0.5)$

$$\Rightarrow V_1 + V_2 + V_3 = 110V \dots (A)$$

Also $V_1 + V_2 = 60 \Rightarrow V_3 = 110 - 60 = 50V$

Now $V_2 + V_3 = 90 \Rightarrow V_2 = 40V$

$$\therefore R_2 = \frac{40}{0.5} = 80\Omega$$



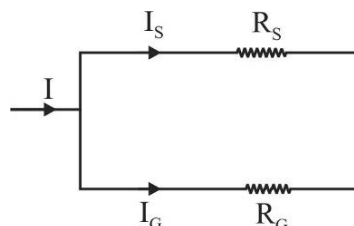
78.(0.75) For ammeter conversion shunt should be connected in parallel

$$V_S = V_G$$

$$I_S R_S = I_G R_G$$

$$\frac{I_S}{I_G} = \frac{4}{3} = \frac{I_S^2 R_S}{H_G^2 R_G} = \left(\frac{R_G}{R_S} \right)^2 \times \frac{R_S}{R_G} = \frac{R_G}{R_S}$$

$$R = \frac{3}{4} R_G$$



79.(243) $R_A = \frac{15.5}{15+5} = 3.75\Omega$

$$(7 + R) \times 0.08 = 20$$

$$R = 243\Omega$$

80.(2) Here in potentiometer. Internal resistance of a cell,

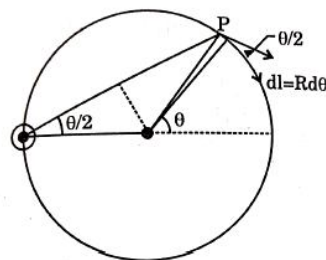
$$r = \frac{\ell_1 - \ell_2}{\ell_2} \times 2\Omega, \text{ Where } \ell_1 = 240 \text{ cm, } \ell_2 = 120 \text{ cm,}$$

$$\therefore r = \frac{240 - 120}{120} \times 2 = \frac{120}{120} \times 2 = 2\Omega$$

81.(8) B at point P = $\frac{\mu_0 i}{2\pi (2R) \cos \frac{\theta}{2}}$

$$\therefore B \cdot dl = \frac{\mu_0 i}{2\pi (2R)} \frac{\cos\left(\frac{\theta}{2}\right) R d\theta}{\cos\left(\frac{\theta}{2}\right)} = \frac{\mu_0 i d\theta}{4\pi}$$

$$\therefore \int_{DEF} B \cdot dl = \int_0^{\pi/2} \frac{\mu_0 i}{4\pi} d\theta = \frac{\mu_0 i}{8}$$



82.(2) $F_m = qvB = \frac{mv^2}{r}$

$$\phi = 2B\pi R^2 + B\pi r^2$$

$$B = \mu_0 nI = \mu_0 nkt$$

$$\phi = B\pi [2R^2 + r^2]$$

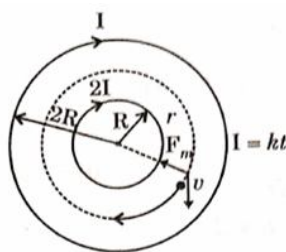
$$E 2\pi r = \frac{d\phi}{dt} = r(2R^2 + r^2) \frac{dB}{dt}$$

$$E = \left(\frac{2R^2 + r^2}{2r} \right) k\mu_0 n$$

$$v = \frac{qEt}{m}$$

From equation (i), $v = \frac{qrB}{m}$

Putting in equation (ii)



$$\frac{qr}{m} B = \frac{qEt}{m}$$

$$r\mu_0 nkt = \left(\frac{2R^2 + r^2}{2r} \right) k\mu_0 nt$$

$$2r^2 = 2R^2 + r^2$$

$$r^2 = 2R^2$$

$$r = \sqrt{2}R$$

$$\frac{r\sqrt{2}}{R} = (\sqrt{2}R) \frac{\sqrt{2}}{R} = 2$$

83.(5) $C_0 = \frac{\epsilon_0 A}{d}$ without metallic slab

$$C' = \frac{\epsilon_0 A}{(Ad)} = 2.5 C_0 \text{ with slab} = 50 \text{ nF}$$

$$W_{\text{ext}} + W_b = U_f - U_i$$

$$\Delta q \text{ (flows into battery)} = (C' - C_0) V = 3 \times 10^{-6}$$

$$\text{So } W_b = -\Delta q \times V = -300 \times 10^{-6}$$

$$W_{\text{ext}} - 300 \times 10^{-6} = \frac{1}{2} (20 - 50) \times 10^{-9} \times (100)^2$$

$$W_{\text{ext}} = +150 \mu J$$

84.(3.8) $E = 1500 \frac{d}{dt} (\mu_0 ni) \times \pi \times \left(\frac{4}{100} \right)^2$

$$= 1500 \times 4\pi \times 10^{-7} \times 240 \times 10^2 \times \frac{20}{0.1} \times \pi \times \frac{4 \times 4}{10^4} \approx 46V$$

$$i = \frac{E}{12} = 3.8A$$

85.(2) Let A & B are boys and ACB is the wire.

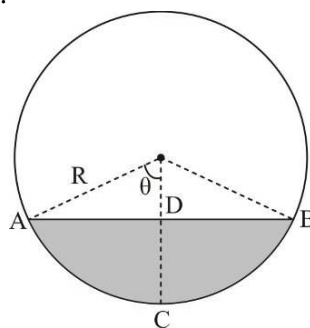
$$AD = \frac{AB}{2} = \frac{4}{2} = 2m$$

$$CD = 1m$$

$$\text{In } \triangle OAD; R^2 = (R-1)^2 + 2^2$$

$$\Rightarrow R = \frac{5}{2}m$$

$$\sin \theta = \frac{2}{2.5} \Rightarrow \theta \approx 53^\circ$$



$$\text{Area of sector } OACB = \frac{106}{360} \times \pi R^2 = \frac{106}{360} \times 3.14 \times \frac{25}{4} = 5.8m^2$$

$$\text{Area of } \triangle OAB = \frac{1}{2} \times 4 \times \frac{3}{2} = 3m^2$$

$$\therefore \text{Area of } ACBDA = 5.8 - 3 = 2.8m^2$$

Angular speed of rotation

$$\omega = 4 \times 2\pi = 8\pi \text{ rad s}^{-1}$$

Flux through area A at any instant can be written as

$$\phi = BA \cos(\omega t + \delta)$$

$$\therefore \text{Induced emf : } E = -\frac{d\phi}{dt} \Rightarrow E = BA\omega \sin(\omega t + \delta)$$

$$E_{rms} = \frac{1}{\sqrt{2}} BA\omega = \frac{1}{\sqrt{2}} \times 4 \times 10^{-5} \times 2.8 \times 8\pi = 201 \times 10^{-5} \text{ V} = 2 \text{ mV}$$

SECTION-6

SUBJECTIVE TYPE

$$86. \left(\sqrt{\frac{2mV_0}{qB_0}} \right)$$

Let after time t , the coordinate of particle is (x, y) and its velocity is $(V_x \hat{i} + V_y \hat{j})$

$$\begin{aligned} \vec{F}_m &= q(V_x \hat{i} + V_y \hat{j}) \times (B_0 x \hat{k}) \\ &= -qV_x B_0 x \hat{j} + qV_y B_0 x \hat{i} \end{aligned}$$

$$\therefore a_x = \frac{qV_y B_0 x}{m}, a_y = -\frac{qV_x B_0 x}{m}$$

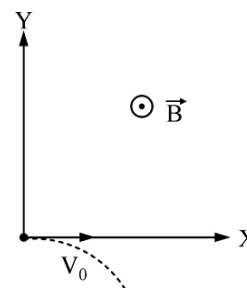
$$V_x \frac{dV_x}{dx} = \frac{qV_y B_0 x}{m}$$

$$\text{Also } V_x^2 + V_y^2 = V_0^2 = \text{const}$$

$$\Rightarrow V_x dV_x = -V_y dV_y \quad \therefore -\frac{V_y dV_y}{dx} = \frac{qV_y B_0 x}{m} \Rightarrow -dV_y = \frac{qB_0}{m} x dx$$

$$\text{When } x - \text{coordinate is maximum, } V_x = 0 \Rightarrow V_y = -V_0$$

$$\therefore -\int_0^{-V_0} dV_y = \frac{qB_0}{m} \int_0^{x_{\max}} x dx \Rightarrow V_0 = \frac{qB_0}{m} \frac{x_{\max}^2}{2} \Rightarrow x_{\max} = \sqrt{\frac{2mV_0}{qB_0}}$$



$$87. \frac{\epsilon_0 A}{d} \left[1 - \frac{\ell \theta}{2d} \right]$$

$$\text{Capacitance of elemental cap is } dC = \frac{\epsilon_0 (b dx)}{(d + x\theta)}$$

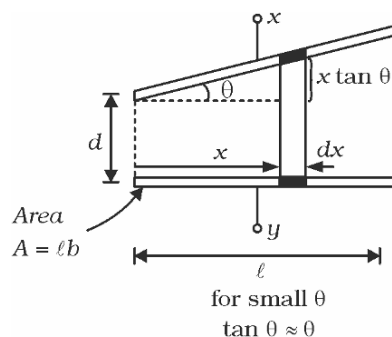
$$\text{Capacitance of this cap is } C = \int dC = \int_0^\ell \frac{\epsilon_0 b dx}{d + x\theta}$$

$$= \frac{\epsilon_0 b}{d} \int_0^\ell \left[1 + \left(\frac{x\theta}{d} \right) \right]^{-1} dx$$

$$= \frac{\epsilon_0 b}{d} \int_0^\ell \left(1 - \frac{x\theta}{d} \right) dx$$

$$= \frac{\epsilon_0 b}{d} \left[x - \frac{\theta}{d} \frac{x^2}{2} \right]_0^\ell$$

$$= \frac{\epsilon_0 b}{d} \left[\ell - \frac{\ell^2 \theta}{2d} \right] = \frac{\epsilon_0 A}{d} \left[1 - \frac{\ell \theta}{2d} \right]$$



$$88. \left(\frac{8\pi \epsilon_0 R K}{K + 1} \right)$$

If a charge q is Supplied to sphere, electric field in surrounding of sphere is given as

Outside dielectric $E_{out} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{x^2}$

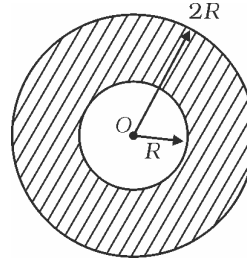
inside dielectric $E_{in} = \frac{1}{4\pi\epsilon_0\epsilon_r} \cdot \frac{q}{x^2}$

Potential of sphere is $V = -\int_{\infty}^R \vec{E} \cdot d\vec{x}$

$$= -\int_{\infty}^{2R} \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{x^2} - \int_{2R}^R \frac{1}{4\pi\epsilon_0\epsilon_r} \cdot \frac{q}{x^2}$$

$$= \frac{q}{4\pi\epsilon_0} \cdot \left[\frac{1}{x} \right]_{\infty}^{2R} + \frac{q}{4\pi\epsilon_0\epsilon_r} \cdot \left[\frac{1}{x} \right]_{2R}^R = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{2R} \right] + \frac{q}{4\pi\epsilon_0\epsilon_r} \left[\frac{1}{2R} \right]$$

$$= \frac{q}{8\pi\epsilon_0 R} \left(1 + \frac{1}{\epsilon_r} \right)$$



89. $\left(R_{AB} = \frac{R}{2}, R_{AC} = \frac{3R}{8} \right)$

From superposition, current in branch AB

$$= \frac{I}{4} + \frac{I}{4} = \frac{I}{2}$$

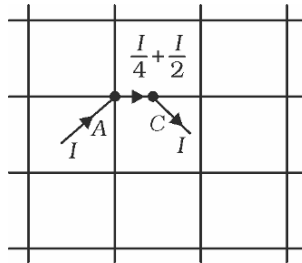
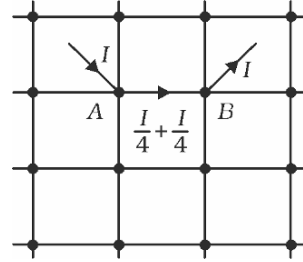
$$\therefore \frac{I}{2} \times R = IR_{AB}$$

$$\Rightarrow R_{AB} = \frac{R}{2}$$

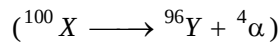
Current in branch AC $= \frac{I}{4} + \frac{I}{2} = \frac{3I}{4}$

$$\therefore \frac{3I}{4} \times \frac{R}{2} = IR_{AC}$$

$$\Rightarrow R_{AC} = \frac{3R}{8}$$



90. $\left(\frac{96\pi m}{25eB} \right)$



The daughter atom will have charge $-2e$ and so they move on circular paths in opposite sense

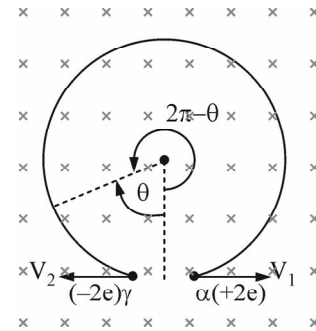
$$R = \frac{mV}{Bq}$$

As momentum of both is same, so R is same

$$\omega_{\alpha} = \frac{Bq}{m_{\alpha}}, \quad \omega_y = \frac{Bq}{m_y}$$

$$\Rightarrow \omega_{rel} = Bq \left(\frac{1}{4m} + \frac{1}{96m} \right) = \frac{B(2e)(100)}{(4)(96m)}$$

$$\therefore t = \frac{2\pi}{\omega} = \frac{96\pi m}{25eB}$$



91.
$$\left(E = \frac{\pi R \sqrt{LC} V^2}{R^2 + 2.25 \left(\frac{L}{C} \right)} \right)$$

Resonance angular frequency $\omega_0 = \frac{1}{\sqrt{LC}}$

When $\omega = 2\omega_0 = \frac{2}{\sqrt{LC}}$ then $X_L = L\omega = \frac{2L}{\sqrt{LC}} = 2\sqrt{\frac{L}{C}}$; $X_C = \frac{1}{\omega C} = \frac{\sqrt{LC}}{2C} = \frac{1}{2}\sqrt{\frac{L}{C}}$

Impedances at frequency 2ω will be $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$$Z^2 = R^2 + \left(2 - \frac{1}{2} \right)^2 \frac{L}{C} = R^2 + 2.25 \frac{L}{C}$$

Power supplied is : $P = VI \cos \phi = \frac{V^2}{Z} \cos \phi = \frac{V^2}{Z} \cdot \frac{R}{Z} = \frac{V^2 R}{Z^2}$

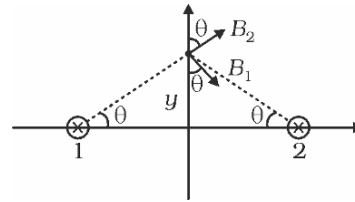
$$P = \frac{V^2 R}{R + 2.25 \frac{L}{C}}$$

$\therefore$ Energy supplied is : $E = PT = P \frac{2\pi}{\omega} = \pi \sqrt{LC} P \Rightarrow E = \frac{\pi R \sqrt{LC} V^2}{R^2 + 2.25 \left(\frac{L}{C} \right)}$

92.
$$\left(\frac{\mu_0 I}{2\pi \ell}, \frac{\mu_0 I^2}{2\pi \ell} \right)$$

$B = 2B_1 \sin \theta$

$$= 2 \frac{\mu_0 I}{2\pi \sqrt{\ell^2 + y^2}} \frac{y}{\sqrt{\ell^2 + y^2}} = \frac{\mu_0 I}{\pi} \frac{y}{\ell^2 + y^2}$$



For $B_{\max}$, $\frac{d}{dy} \left[\frac{y}{\ell^2 + y^2} \right] = 0 \Rightarrow (\ell^2 + y^2)(1) - y(2y) = 0 \Rightarrow y = \pm \ell$

$\therefore B_{\max} = \frac{\mu_0 I}{\pi} \frac{\ell}{2\ell^2} = \frac{\mu_0 I}{2\pi \ell}$

$\therefore$ Maximum force on the third wire of unit length $F_{\max} = I(1)B_{\max} = \frac{\mu_0 I^2}{2\pi \ell}$

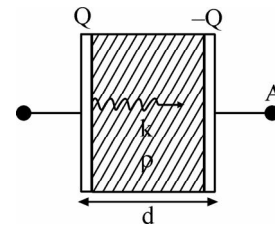
93.
$$\left[\rho k \epsilon_0, \frac{Q_0}{RC} e^{\frac{-t}{\rho k \epsilon_0}} \right]$$

$C = \frac{k \epsilon_0 A}{d}, R = \frac{\rho d}{A}$

Charge flows from positive to negative plate through the dielectric

$\tau = RC = \rho k \epsilon_0$

Leakage current $I =$ discharging current $= \frac{Q_0}{RC} e^{\frac{-t}{\tau}} = \frac{Q_0}{RC} e^{\frac{-t}{\rho k \epsilon_0}}$



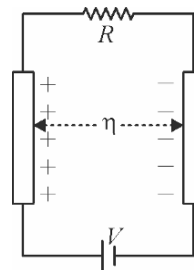
94.

$$\sqrt{\frac{\mu_0}{\epsilon_0} \frac{\ell n \eta}{\pi}}$$

Figure shows the situation described in the question. The two wires form a wire capacitor of which capacitance is given as $C = \frac{\pi \epsilon_0 \ell}{\ell n \eta}$

Charge on the capacitor in steady state is given as

$$q = CV = \frac{\pi \epsilon_0 \ell V}{\ell n \eta} \quad \dots (1)$$



Due to voltage source a constant current flows in the wires which repel each other by a magnetic force which is nullified by the force of attraction on the two wires due to opposite charges on the two wires as shown in figure. The force of attraction is given as

$$F_1 = qE = q \left(\frac{q}{2\pi \epsilon_0 \ell (\eta r)} \right) \quad \dots (2)$$

Substituting the value of q from equation (1) into (2),

$$F_1 = \frac{\left(\frac{\pi \epsilon_0 \ell V}{\ell n \eta} \right)^2}{2\pi \epsilon_0 \ell (\eta r)} \quad \dots (3)$$

The current flow is in the opposite direction in the two wires so the net force of repulsion between these wires is given as

$$F_2 = \frac{\mu_0 I^2 \ell}{2\pi (\eta r)} = \frac{\mu_0 (V/R)^2 \ell}{2\pi \eta r} \quad \dots (4)$$

As the net force between the two wires is zero we use

$$\frac{\mu_0 (V/R)^2 \ell}{2\pi \eta r} = \frac{\left(\frac{\pi \epsilon_0 \ell V^2}{\ell n \eta} \right)^2}{2\pi \epsilon_0 \ell (\eta r)} \Rightarrow R = \sqrt{\frac{\mu_0}{\epsilon_0} \frac{\ell n \eta}{\pi}}$$

95.

$$2x \sqrt{\frac{4 \epsilon_0 I}{P \sigma}}$$

Due to cup, EF at the centre is $E = \frac{\sigma}{4 \epsilon_0}$

For small angular displacement of dipole, torque on it is $\tau = PE \sin \theta \approx PE \theta$

Restoring angular acceleration of dipole $\alpha = -\frac{\tau}{I} = -\frac{PE}{I} \cdot \theta$ (i)

For SHM we use $\alpha = -\omega^2 \theta$ (ii)

Comparing (i) & (ii) $\omega = \sqrt{\frac{PE}{I}} = \sqrt{\frac{P \sigma}{4 \epsilon_0 I}}$

Time period of oscillation $T = \frac{2\pi}{\omega} = 2x \sqrt{\frac{4 \epsilon_0 I}{P \sigma}}$

96. (i) $[24 \cos(5 \times 10^4 t) \mu F]$ (ii) $(10.5 \mu s)$

(i) In the steady state C_1 and C_2 are in series arrangement. Their equivalent capacitance

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = 1.2 \mu F$$

Charge on the capacitor C_2 , $Q_0 = C_{eq} V = 1.2 \times 20 = 24 \mu F$

When S_1 is open and S_2 is closed, capacitor C_2 starts discharging through the inductor and let at any time 't' charge on the capacitor be Q , then applying Kirchoff's voltage law,

$$\frac{Q}{C_2} - L \frac{dI}{dt} = 0$$

$$\frac{Q}{C_2} + L \frac{d^2 Q}{dt^2} = 0 \quad \left[I = -\frac{dQ}{dt} \right]$$

$$\Rightarrow Q = Q_0 \sin(\omega t + \phi)$$

$$\text{Where } \omega = \frac{1}{\sqrt{LC_2}} = 5 \times 10^4 \text{ rad/s}$$

$$\text{At } t = 0 \quad Q = Q_0$$

$$\text{Hence } \phi = \frac{\pi}{2}$$

Thus charge on the capacitor at any time t is $Q = Q_0 \cos \omega t$

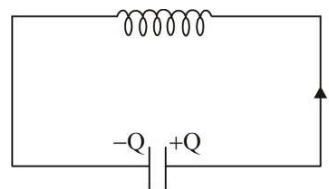
(ii) In a LC circuit, total energy is always conserved

$$\therefore U_L + U_C = \frac{Q_0^2}{2C_2}$$

$$\text{At the time } t = t_1, U_L = \frac{1}{3} U_C$$

$$\text{Hence } U_C = \frac{3}{4} \left(\frac{1}{2} \frac{Q_0^2}{C_2} \right) \Rightarrow \frac{1}{2} \frac{Q^2}{C_2} = \frac{3}{4} \left(\frac{1}{2} \frac{Q_0^2}{C_2} \right)$$

$$Q = \frac{\sqrt{3}}{2} Q_0 \text{ or } Q_0 \cos \omega t_1 = \frac{\sqrt{3}}{2} Q_0; \quad \omega t_1 = \frac{\pi}{6} \text{ or } t_1 = \frac{\pi}{6\omega} = 1.05 \times 10^{-5} = 10.5 \mu s$$



$$97. (i) \quad \boxed{\varepsilon = N \pi a^2 B_0 \omega \sin \omega t, \langle P \rangle = \frac{(N \pi a^2 B_0 \omega)^2}{2R}} \quad (ii) \quad \boxed{R = \frac{\mu_0 N^2 \pi a \omega}{4 \tan \theta}}$$

(i) After some time t , the normal to the coil plane makes an angle ωt with the magnetic field $\vec{B} = B_0 \vec{i}$.

Then, the magnetic flux through the coil is $\phi = N \vec{B} \cdot \vec{S}$

where the vector surface $\vec{S}$ is given by $\vec{S} = \pi a^2 (\cos \omega t \vec{i} + \sin \omega t \vec{j})$

$$\text{Therefore } \phi = N \pi a^2 B_0 \cos \omega t$$

$$\text{The induced electromotive force is } \varepsilon = -\frac{d\phi}{dt} \Rightarrow \boxed{\varepsilon = N \pi a^2 B_0 \omega \sin \omega t}$$

$$\text{The instantaneous power is } P = \varepsilon^2 / R, \text{ therefore } \boxed{\langle P \rangle = \frac{(N \pi a^2 B_0 \omega)^2}{2R}}$$

$$\text{where we used } \langle \sin^2 \omega t \rangle = \frac{1}{T} \int_0^T \sin^2 \omega t dt = \frac{1}{2}$$

(ii) The total field at the center the coil at the instant t is $\vec{B}_t = \vec{B}_0 + \vec{B}_i$

where $\vec{B}_i$ is the magnetic field due to the induced current $\vec{B}_i = B_i (\cos \omega t \vec{i} + \sin \omega t \vec{j})$

with $B_i = \frac{\mu_0 N I}{2a}$ and $I = \varepsilon / R$ Therefore $B_i = \frac{\mu_0 N^2 \pi a B_0 \omega}{2R} \sin \omega t$

The mean values of its components are

$$\langle B_{ix} \rangle = \frac{\mu_0 N^2 \pi a B_0 \omega}{2R} \langle \sin \omega t \cos \omega t \rangle = 0$$

$$\langle B_{iy} \rangle = \frac{\mu_0 N^2 \pi a B_0 \omega}{2R} \langle \sin^2 \omega t \rangle = \frac{\mu_0 N^2 \pi a B_0 \omega}{4R}$$

And the mean value of the total magnetic field is $\langle \vec{B}_t \rangle = B_0 \vec{i} + \frac{\mu_0 N^2 \pi a B_0 \omega}{4R} \vec{j}$

The needle orients along the mean field, therefore $\tan \theta = \frac{\mu_0 N^2 \pi a \omega}{4R}$

Finally, the resistance of the coil measured by this procedure, in terms of θ , is

$$R = \frac{\mu_0 N^2 \pi a \omega}{4 \tan \theta}$$

98. (i) $\boxed{\frac{5}{3}}$

(ii) $\boxed{E_m}$

(iii) $\boxed{\sqrt{\frac{7}{3}}}$

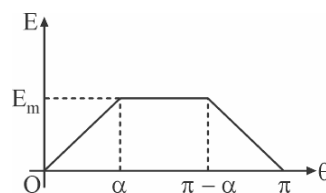
(i) Output waveform is as shown

$$\text{Area under curve} = \frac{1}{2}(\pi + \pi - 2\alpha)E_m$$

$$\text{So, average value} = \frac{2(\pi - \alpha)}{2\pi} E_m = \left(\frac{\pi - \alpha}{\pi}\right) E_m$$

$$\text{When } \alpha = \frac{\pi}{6} ; E_{av} = \left(\frac{\pi - \pi/6}{\pi}\right) E_m = \frac{5}{6} E_m$$

$$\text{When } \alpha = \frac{\pi}{2} ; E_{av} = \left(\frac{\pi - \pi/2}{\pi}\right) E_m = \frac{E_m}{2}$$



$$\therefore \text{Ratio} = \frac{\frac{5}{6} E_m}{\frac{E_m}{2}} = \frac{5}{6} \times \frac{2}{1} = \frac{5}{3}$$

$$(ii) E_{rms} = \sqrt{\left\{ \frac{1}{\pi} \left[2 \int_0^\alpha \frac{E_m^2}{\alpha^2} \theta^2 d\theta + \int_\alpha^{\pi-\alpha} E_m^2 d\theta \right] \right\}} = \left[\frac{E_m^2}{\pi} \left(2 \left(\frac{\theta^3}{3\alpha^2} \right)_0^\alpha + (\theta)_\alpha^{\pi-\alpha} \right) \right]^{1/2} = E_m \sqrt{\frac{1}{\pi} \left(\pi - \frac{4\alpha}{3} \right)}$$

For $\alpha = 0$, $E_{rms} = E_m$

$$(iii) \text{ For } \alpha = \frac{\pi}{6} ; E_{rms} = E_m \sqrt{\left(\frac{\pi - \frac{4}{3} \cdot \frac{\pi}{6}}{\pi} \right)} = \sqrt{\frac{7}{9}} E_m$$

$$\text{For } \alpha = \frac{\pi}{2} ; E_{rms} = \frac{E_m}{\sqrt{3}} \therefore \text{Ratio} = \sqrt{\left(\frac{7}{9} \times \frac{3}{1} \right)} = \sqrt{\frac{7}{3}}$$

99. (i) $\boxed{5C}$

(ii) $\boxed{50 \text{ kA}}$

(iii) $\boxed{10h}$

(i) The total charge Q is just the area under the curve of the figure. Because of the triangular shape, we

immediately get : $Q = \frac{I_0 \tau}{2} = 5C$.

(ii) The average current is $I = Q / \tau = \frac{I_0}{2} = 50 \text{ kA}$, . Simply half the maximal value.

- (iii) Since the bottom of the cloud gets negatively charged and the ground positively charged, the situation is essentially that of a giant parallel-plate capacitor. The amount of energy stored just before lightning occurs is $QE_0h/2$ where E_0h is the voltage difference between the bottom of the cloud and the ground, and lightning releases this energy. Altogether we thus get for one lightning the energy $QE_0h/2 = 7.5 \times 10^8 J$. It follows that you could light up the 100W bulb for the duration

$$t = \frac{32 \times 10^6}{6.5 \times 10^9} \times \frac{7.5 \times 10^8 J}{100W} \approx 10h.$$

100. (i) (a) $\boxed{\frac{\epsilon_0 AV}{d-x}, \frac{\epsilon_0 AV}{d+x}}$ (b) $\boxed{\frac{2\epsilon_0 AV^2}{d^3} x}$ (c) $\boxed{-\frac{2}{m} \left(k - \frac{\epsilon_0 AV^2}{d^3} \right) d}$

(ii) $\boxed{V \frac{2\epsilon_0 Ax}{d^2 C_s + 2\epsilon_0 Ad}}$

- (i) (a) For the given value of x , the amount of charge on each capacitor is

$$Q_1 = VC_1 = \frac{\epsilon_0 AV}{d-x} \quad \text{and} \quad Q_2 = VC_2 = \frac{\epsilon_0 AV}{d+x}$$

- (b) Note that we have two capacitors. By using the formula for force between plates of a capacitor, we get $F_1 = \frac{Q_1^2}{2\epsilon_0 A}$, $F_2 = \frac{Q_2^2}{2\epsilon_0 A}$

As these two forces are in the opposite directions, the net electric force is

$$F_E = F_1 - F_2 \Rightarrow F_E = \frac{\epsilon_0 AV^2}{2} \left(\frac{1}{(d-x)^2} - \frac{1}{(d+x)^2} \right)$$

Ignoring terms of order x^2 in the answer we get $F_E = \frac{2\epsilon_0 AV^2}{d^3} x$

- (c) There are two springs placed in series with the same spring constant, k , then the mechanical force is $F_m = -2kx$

Combining this result and noticing that these two forces are in the opposite directions, we

$$\text{get : } F = F_m + F_E \Rightarrow F = -2 \left(k - \frac{\epsilon_0 AV^2}{d^3} \right) x,$$

By using the Newton's second law, $F = ma$.

$$\text{And the answer, we get : } a = -\frac{2}{m} \left(k - \frac{\epsilon_0 AV^2}{d^3} \right) d$$

- (ii) Starting with Kirchhoff's laws, for two electric circuits, we have
- $$\begin{cases} \frac{Q_s}{C_s} + V - \frac{Q_2}{C_2} = 0 \\ -\frac{Q_s}{C_s} + V - \frac{Q_1}{C_1} = 0 \\ Q_2 - Q_1 + Q_s = 0 \end{cases}$$

$$\text{Noting that } V_s = \frac{Q_s}{C_s} \text{ one obtains} \Rightarrow V_s = \frac{\frac{2\epsilon_0 Ax}{d^2 - x^2}}{C_s + \frac{2\epsilon_0 Ad}{d^2 - x^2}}.$$

$$\text{Ignoring terms of order } x^2 \text{ in the answer we get : } V_s = V \frac{2\epsilon_0 Ax}{d^2 C_s + 2\epsilon_0 Ad}$$